

Requirements Checklist

Module: **CT050-3-2-WAPP – Web Applications** (Group Assignment)

This checklist is the traceability baseline for the project. Every requirement has a stable ID. The final audit ([REQUIREMENT_TRACEABILITY.md](#)) maps each ID to its implementation, evidence and status.

Legend: **M** = mandatory (assignment brief), **O** = optional improvement.  = implemented and verified against the code; **PENDING** = built and documented, but the hosting step itself has not been performed yet (see §14).

Every requirement below was re-checked against the code on the `feature/railway-firebase-sqlite` branch. The measured evidence behind the ticks is **211 automated tests passing, 42 browser checks passing** (14 Playwright tests x desktop, tablet and mobile) and a seeded catalogue of **8 categories, 12 courses, 64 learning resources, 11 quizzes, 54 questions and 216 answer options**. File-level implementation and evidence for each ID are in [REQUIREMENT_TRACEABILITY.md](#).

1. Technology and platform

ID	Requirement	Type	Status
TECH-01	Built with .NET technologies – ASP.NET Core MVC on the latest stable LTS (.NET 10)	M	
TECH-02	C# for all server-side code	M	
TECH-03	Database connectivity through Entity Framework Core	M	
TECH-04	One database provider (SQLite) with code-first EF Core migrations	M	
TECH-05	ASP.NET Core Identity for authentication	M	
TECH-06	Razor views, HTML5 semantic markup	M	
TECH-07	CSS3 with Bootstrap 5.3 customised by a project design system	M	
TECH-08	JavaScript for client-side processing where appropriate	M	
TECH-09	Multimedia (images, embedded video, PDF documents, SVG illustrations)	M	
TECH-10	Server-side processing (controllers, services, EF Core queries)	M	
TECH-11	Proper project and file organisation	M	

2. Guest (anonymous visitor)

ID	Requirement	Type	Status
GUEST-01	View Home page	M	
GUEST-02	View About page	M	

ID	Requirement	Type	Status
GUEST-03	Browse published courses	M	✓
GUEST-04	Search courses (server-side)	M	✓
GUEST-05	Filter courses by category and difficulty	M	✓
GUEST-06	Open course details	M	✓
GUEST-07	View selected public (preview) learning resources	M	✓
GUEST-08	Register	M	✓
GUEST-09	Login	M	✓
GUEST-10	View Contact page and send a contact message	M	✓

3. Registered user (Student)

ID	Requirement	Type	Status
STU-01	Student dashboard with meaningful, data-driven information	M	✓
STU-02	Dashboard: enrolled courses, completed courses, available courses	M	✓
STU-03	Dashboard: recent activity, quiz results, learning progress	M	✓
STU-04	View and update profile	M	✓
STU-05	Browse courses and view course details	M	✓
STU-06	Enrol in a course (and leave a course)	M	✓
STU-07	View My Courses	M	✓
STU-08	Access learning resources of enrolled courses	M	✓
STU-09	Track learning progress (mark resources complete)	M	✓
STU-10	Take quizzes	M	✓
STU-11	View quiz results and history	M	✓
STU-12	Logout	M	✓
STU-13	Change password	O	✓

4. Administrator

ID	Requirement	Type	Status
ADM-01	Separate, protected Admin area	M	✓
ADM-02	Admin dashboard with real platform statistics	M	✓

ID	Requirement	Type	Status
ADM-03	Courses CRUD (create, list, details, edit, delete)	M	✓
ADM-04	Categories CRUD	M	✓
ADM-05	Learning resources CRUD (incl. secure file upload)	M	✓
ADM-06	Enrolments management (list, enrol a student, remove)	M	✓
ADM-07	Quizzes CRUD	M	✓
ADM-08	Quiz questions and answer options CRUD, set correct answer	M	✓
ADM-09	Quiz results (attempts) view and delete	M	✓
ADM-10	Users: list, view details, change role, deactivate/reactivate, safe delete	M	✓
ADM-11	Website content: contact message inbox	O	✓

5. Database

ID	Requirement	Type	Status
DB-01	Relational model with primary keys and foreign keys	M	✓
DB-02	Required properties and maximum lengths	M	✓
DB-03	Unique constraints where appropriate	M	✓
DB-04	Indexes where useful	M	✓
DB-05	Intentional cascade / restrict delete behaviour	M	✓
DB-06	CreatedAt / UpdatedAt timestamps where useful	M	✓
DB-07	EF Core migrations reproducible from an empty database	M	✓
DB-08	Realistic seed data (8 categories, 12 courses, 64 resources, 11 quizzes, demo activity)	M	✓
DB-09	Insert, Display, Update, Delete operations demonstrated	M	✓
DB-10	ERD that exactly matches the EF Core entities	M	✓

6. Authentication and authorisation

ID	Requirement	Type	Status
AUTH-01	Register, Login, Logout	M	✓
AUTH-02	Password hashing through Identity (no plain-text passwords)	M	✓
AUTH-03	Roles: Guest, Student, Admin with role-based authorisation	M	✓
AUTH-04	Protected pages for students and admin-only controllers	M	✓

ID	Requirement	Type	Status
AUTH-05	Initial admin seeded from secrets / environment variables (no password in Git)	M	✓
AUTH-06	Account lockout after repeated failed logins	O	✓

7. Validation

ID	Requirement	Type	Status
VAL-01	Server-side validation with DataAnnotations and ModelState	M	✓
VAL-02	Client-side validation with ASP.NET validation helpers	M	✓
VAL-03	Bootstrap-styled, accessible validation feedback	M	✓
VAL-04	Required, email, password, length and range rules	M	✓
VAL-05	File validation (type, size, content signature)	M	✓
VAL-06	Business-rule validation (e.g. exactly one correct answer)	M	✓

8. Security

ID	Requirement	Type	Status
SEC-01	No SQL injection (EF Core parameterised queries only)	M	✓
SEC-02	XSS protection (Razor encoding, no raw user HTML, CSP)	M	✓
SEC-03	CSRF protection (anti-forgery tokens on all unsafe requests)	M	✓
SEC-04	IDOR protection (ownership checks on attempts, resources, files)	M	✓
SEC-05	Overposting protection (ViewModels, never bind entities)	M	✓
SEC-06	Safe file uploads (allow-list, magic bytes, size limit, random names, outside wwwroot)	M	✓
SEC-07	Safe video embedding (strict YouTube/Vimeo parsing, frame-src allow-list)	M	✓
SEC-08	No secrets committed to Git	M	✓
SEC-09	Admin routes protected by role authorisation	M	✓
SEC-10	Security headers, HTTPS/HSTS in production	O	✓
SEC-11	Rate limiting on login, registration and contact form	O	✓

9. Error handling and logging

ID	Requirement	Type	Status
ERR-01	Custom 404 page	M	✓

ID	Requirement	Type	Status
ERR-02	Custom 500 / error page without stack traces in production	M	✓
ERR-03	User-friendly validation and business error messages	M	✓
ERR-04	Database error handling (e.g. unique-constraint races)	M	✓
ERR-05	Structured logging of important events	M	✓

10. UI / UX, accessibility and responsiveness

ID	Requirement	Type	Status
UI-01	Coherent custom design system (not default Bootstrap)	M	✓
UI-02	Navigation bar per role, hero, course cards, dashboards, tables, forms, footer	M	✓
UI-03	Empty states and success/error messages	M	✓
UI-04	Breadcrumbs where useful	M	✓
UI-05	Responsive on mobile, tablet and desktop; collapsing navigation; usable tables	M	✓
UI-06	Semantic HTML, labels, alt text, keyboard access, contrast, heading structure	M	✓
UI-07	No placeholder/template text, no broken images or dead links	M	✓

11. Testing and quality

ID	Requirement	Type	Status
TEST-01	Automated unit tests for services and helpers	M	✓
TEST-02	Automated integration tests (auth, protected routes, CRUD, enrolment, quiz, search, validation, 404)	M	✓
TEST-03	Browser end-to-end tests across mobile, tablet and desktop viewports	O	✓
TEST-04	Manual testing checklist with recorded results	M	✓
TEST-05	Build without errors and without warnings	M	✓

12. Repository, CI/CD and deployment

ID	Requirement	Type	Status
GIT-01	Git repository with a professional structure and <code>.gitignore</code>	M	✓

ID	Requirement	Type	Status
GIT-02	Sensible commit history and documented branching strategy	M	✓
GIT-03	README with all required sections	M	✓
CI-01	GitHub Actions CI: restore, build, test, fail on error	M	✓
CI-02	GitHub Actions container workflow that reproduces the Railway runtime	M	✓
DEP-01	Production ASP.NET application packaged and ready to run on Railway	M	PENDING
DEP-02	SQLite production database on a persistent volume, configured through environment variables	M	PENDING
DEP-03	Documented migration strategy and production verification	M	✓
PAGES-01	Separate static presentation site with an "Open Learning System" call to action	M	✓
PAGES-02	Presentation site deployment automated and reproducible	M	✓

13. Documentation (university deliverables)

ID	Requirement	Type	Status
DOC-01	Proposal: title, objectives, mission, audience modelling, scope	M	✓
DOC-02	Final report content (full structure from the brief)	M	✓
DOC-03	ERD (Mermaid)	M	✓
DOC-04	Use cases	M	✓
DOC-05	Flowcharts (registration, login, enrolment, quiz, admin course CRUD)	M	✓
DOC-06	Wireframes	M	✓
DOC-07	Navigation structure diagram	M	✓
DOC-08	Testing documentation	M	✓
DOC-09	Deployment documentation	M	✓
DOC-10	Team responsibilities and Git workflow	M	✓
DOC-11	Viva preparation questions	M	✓

14. Notes on partial or pending items

Everything above is either verified in the code or listed here. No requirement is claimed that the code does not support.

ID	Note
DEP-01, DEP-02	The Docker image, <code>railway.json</code> , the environment-variable contract and <code>scripts/set-app-url.sh</code> are complete, and the <code>container</code> CI job runs the image the way Railway does, including a restart that proves the SQLite file on <code>/data</code> survives. The Railway CLI is installed but not authenticated , so no Railway deployment has been made and no Railway URL is claimed.
ADM-09	The three attempt pages exist and are crawled and scanned by the test suite, but there is no dedicated automated test for deleting a single attempt.
TEST-04	TESTING.md §7 records 13 passed checks. Keyboard-only journeys, a screen-reader pass, real devices and the live production check remain "To do" because they need a person or a deployment.
UI-06	Automated axe-core WCAG 2.2 A/AA scans pass on the public, student and admin pages at all three viewports. This does not replace the manual keyboard and screen-reader pass above.
PAGES-01	The presentation site is live at https://learnhub-wapp.web.app . Its buttons carry the conventional Railway address until <code>scripts/set-app-url.sh</code> rewrites it after the first deployment.
GIT-03	<code>README.md</code> has all required sections, but its badges, hosting table and deployment section still describe the earlier Azure hosting and are not yet updated.

15. Explicitly out of scope

To prevent scope creep, the following are **not** part of LearnHub v1.0: AI tutor, live video classrooms, real-time chat, payment processing, certificates, e-mail delivery (confirmation / password reset e-mails), instructor self-service role, discussion forums, mobile apps, multi-language UI.